
Stochastic Reliability and Risk Simulation for Networked Systems

Using Continuous-Time Markov Chains, Monte Carlo Simulation,
and Variance Reduction Techniques

Ujjwal Karki

M.S. Mathematics (Statistics Concentration)

Department of Mathematics and Statistics

Georgia State University, Atlanta, GA 30303

ukarki2@student.gsu.edu

Date: March 19, 2026

Abstract

This report presents a comprehensive stochastic reliability analysis of a networked system comprising 15 interacting components, modeled using Continuous-Time Markov Chains (CTMCs). Each component is governed by exponentially distributed failure and repair processes with rates λ_i and μ_i derived from real-world reliability benchmarks. We implement large-scale Monte Carlo simulation (100,000 trials) to estimate key system-level metrics including steady-state availability, failure frequency, and mean time to failure (MTTF). System reliability is derived analytically for series, parallel, and k -out-of- n configurations, with simulation results validated against closed-form solutions. We perform sensitivity analysis via closed-form perturbation of transition intensities, apply importance sampling for efficient rare-event probability estimation, and optimize redundancy allocation under a \$5,000 cost constraint using a greedy marginal-gain algorithm. The optimized

configuration achieves a 20.4% improvement in system reliability. Applications of this methodology span nuclear engineering, telecommunications, aerospace, power systems, and financial risk modeling.

All computational work performed in Python (NumPy, SciPy, pandas).

Contents

1	Introduction	3
1.1	Motivation	3
1.2	Research Objectives	3
2	Background and Related Literature	4
2.1	Reliability Theory	4
2.2	Continuous-Time Markov Chains in Reliability	4
2.3	Monte Carlo Methods in Reliability	4
2.4	Redundancy Optimization	5
3	System Model and Notation	5
3.1	Component Model	5
3.2	Parameter Calibration	5
3.3	System Architecture	6
4	Analytical Framework	6
4.1	Steady-State Component Availability	6
4.2	Time-Dependent Availability via Matrix Exponential	7
4.3	System Topology Reliability	7
4.3.1	Series Configuration	7
4.3.2	Parallel Configuration	7
4.3.3	k -out-of- n Configuration	8
5	Monte Carlo Simulation	8
5.1	Simulation Design	8
5.2	Estimated Metrics	9
5.3	Validation	9
6	Sensitivity Analysis	9
6.1	Methodology	9
6.2	Results and Interpretation	10
7	Importance Sampling for Rare-Event Estimation	10
7.1	Motivation	10
7.2	Importance Sampling Estimator	10
7.3	Likelihood Ratio for Poisson Failure Counts	11
7.4	Results	11
8	Redundancy Optimization	11
8.1	Problem Formulation	11

8.2	Greedy Marginal-Gain Algorithm	12
8.3	Component Costs and Optimal Allocation	12
8.4	Results	13
9	Applications of CTMC Reliability Analysis	13
9.1	Nuclear Power and Safety-Critical Systems	13
9.2	Telecommunications and Network Infrastructure	13
9.3	Aerospace and Aviation	13
9.4	Power Grids and Smart Energy Systems	14
9.5	Financial Risk and Operational Resilience	14
9.6	Healthcare and Medical Device Reliability	14
9.7	Software and Cloud Computing Systems	14
9.8	Manufacturing and Supply Chain	14
10	Discussion	15
10.1	Key Findings	15
10.2	Limitations	15
10.3	Future Directions	16
11	Conclusion	16
	Acknowledgments	17
	References	18
A	Python Implementation Notes	19
A.1	Key Library Functions	19
A.2	Reproducibility	19
A.3	Computational Cost	19

1 Introduction

Modern engineered systems — from power grids and aircraft to data centers and financial clearing networks — depend on the reliable operation of many interdependent components. A single component failure can cascade into system-wide outages with enormous economic and safety consequences. Quantifying the probability and impact of such failures is the domain of *reliability engineering*, a field that draws deeply on probability theory, stochastic processes, and statistical simulation.

This project develops a fully integrated computational framework for reliability analysis of a 15-component networked system. The framework combines exact analytical results from Continuous-Time Markov Chain (CTMC) theory with large-scale Monte Carlo simulation, sensitivity analysis, rare-event estimation via importance sampling, and constrained combinatorial optimization for redundancy allocation.

1.1 Motivation

As systems grow in complexity, purely analytical solutions become intractable. The system state space for a 15-component network with binary (working/failed) components has $2^{15} = 32,768$ possible states — a space large enough to demand simulation-based estimation for many metrics of interest. At the same time, events such as simultaneous multi-component failure are so rare that naive Monte Carlo produces zero estimates; specialized variance reduction techniques are essential.

This project bridges the gap between tractable analytical models and high-fidelity stochastic simulation, providing both rigorous theoretical justification and practical numerical implementation.

1.2 Research Objectives

The specific objectives of this analysis are:

- O1.** Model each of 15 system components as a 2-state CTMC with failure rate λ_i and repair rate μ_i calibrated to real-world reliability data.
- O2.** Derive closed-form expressions for per-component steady-state availability, MTTF, and MTTR, and compute time-dependent availability via matrix-exponential solutions.
- O3.** Implement Monte Carlo simulation with 10^5 trials to estimate system-level availability, expected failure frequency, and MTTF under varying component configurations.
- O4.** Derive and compare system reliability under series, parallel, and k -out-of- n structural assumptions; validate simulation estimates against analytical benchmarks.

- O5.** Quantify the marginal sensitivity of system reliability to perturbations in individual component transition rates λ_i and μ_i using closed-form analytical differentiation.
- O6.** Apply importance sampling to efficiently estimate rare-event probabilities (simultaneous multi-component failure) that are inaccessible to standard Monte Carlo.
- O7.** Solve a constrained redundancy allocation problem: maximize system reliability subject to a total equipment budget of \$5,000.

2 Background and Related Literature

2.1 Reliability Theory

Reliability engineering emerged as a formal discipline in the 1950s, driven by the US military's need to quantify the failure behavior of electronic equipment [1]. The *reliability function* $R(t)$ gives the probability that a system operates without failure through time t :

$$R(t) = P(T > t), \quad t \geq 0,$$

where T is the random time to first failure. For systems with exponentially distributed failure times, $R(t) = e^{-\lambda t}$, where λ is the constant failure rate.

2.2 Continuous-Time Markov Chains in Reliability

CTMCs provide the natural mathematical framework for systems with multiple states (working, degraded, failed) and memoryless transition times. A CTMC $\{X(t), t \geq 0\}$ on state space $\mathcal{S}$ is characterized by its *generator matrix* $\mathbf{Q}$, where:

$$q_{ij} = \lim_{\Delta t \rightarrow 0} \frac{P(X(t + \Delta t) = j \mid X(t) = i)}{\Delta t}, \quad i \neq j, \quad q_{ii} = -\sum_{j \neq i} q_{ij}.$$

The transient state distribution satisfies the Kolmogorov forward equation:

$$\frac{d}{dt} \boldsymbol{\pi}(t) = \boldsymbol{\pi}(t) \mathbf{Q}, \quad \boldsymbol{\pi}(t) = \boldsymbol{\pi}(0) e^{\mathbf{Q}t},$$

where $e^{\mathbf{Q}t}$ is the matrix exponential [2].

2.3 Monte Carlo Methods in Reliability

Monte Carlo simulation has been central to reliability analysis since the Manhattan Project. By simulating many independent realizations of system evolution, one estimates quantities whose analytical computation is intractable. For rare-event probabilities, standard Monte Carlo requires $O(1/p)$ samples to achieve relative error of order 1, making it

infeasible when $p \ll 1$. Importance sampling addresses this by simulating under a biased distribution and reweighting [3].

2.4 Redundancy Optimization

Redundancy allocation — adding backup components to critical subsystems — is a classical combinatorial optimization problem. The goal is to maximize system reliability subject to constraints on cost, weight, or space. For series-parallel systems, dynamic programming and greedy heuristics are standard approaches [4].

3 System Model and Notation

3.1 Component Model

We consider a networked system of $n = 15$ components. Each component i ($i = 1, \dots, 15$) is modeled as a 2-state CTMC on state space $\mathcal{S}_i = \{0, 1\}$, where state 0 denotes *working* and state 1 denotes *failed*. The generator matrix for component i is:

$$\mathbf{Q}_i = \begin{pmatrix} -\lambda_i & \lambda_i \\ \mu_i & -\mu_i \end{pmatrix}, \quad (1)$$

where $\lambda_i > 0$ is the failure rate (transitions $0 \rightarrow 1$) and $\mu_i > 0$ is the repair rate (transitions $1 \rightarrow 0$). Both rates are expressed in units of failures (or repairs) per hour.

3.2 Parameter Calibration

Failure and repair rates are calibrated to reflect a realistic industrial setting, consistent with published reliability handbooks for electronic and electromechanical components [7]. Table 1 presents the parameter set used throughout this analysis.

Table 1: Component failure rates (λ_i), repair rates (μ_i), steady-state availability (A_i), MTTF, and MTTR.

Component	λ_i (hr ⁻¹)	μ_i (hr ⁻¹)	A_i	MTTF (hr)	MTTR (hr)
1	0.0020	0.1000	0.9804	500.0	10.00
2	0.0030	0.0800	0.9639	333.3	12.50
3	0.0010	0.1200	0.9917	1000.0	8.33
4	0.0040	0.0900	0.9574	250.0	11.11
5	0.0020	0.1100	0.9821	500.0	9.09
6	0.0050	0.0700	0.9333	200.0	14.29
7	0.0030	0.1000	0.9709	333.3	10.00
8	0.0020	0.0900	0.9783	500.0	11.11
9	0.0010	0.1300	0.9924	1000.0	7.69
10	0.0040	0.0800	0.9524	250.0	12.50
11	0.0030	0.1000	0.9709	333.3	10.00
12	0.0020	0.1100	0.9821	500.0	9.09
13	0.0050	0.0700	0.9333	200.0	14.29
14	0.0010	0.1200	0.9917	1000.0	8.33
15	0.0030	0.0900	0.9677	333.3	11.11
Mean	0.0028	0.0953	0.9699	432.2	10.63

3.3 System Architecture

The 15 components are embedded in a networked topology analyzed under three structural assumptions: *series*, *parallel*, and *k-out-of-n*. The *k-out-of-n* model with $k = 8$ (majority voting) is the primary model of interest, reflecting realistic fault-tolerant system design where a majority of subsystems must operate for the system to function.

4 Analytical Framework

4.1 Steady-State Component Availability

For each 2-state CTMC with generator (1), the steady-state distribution $\boldsymbol{\pi}_i = (\pi_i^{(0)}, \pi_i^{(1)})$ satisfies $\boldsymbol{\pi}_i \mathbf{Q}_i = \mathbf{0}$ with $\pi_i^{(0)} + \pi_i^{(1)} = 1$. Solving gives the steady-state availability:

$$A_i = \pi_i^{(0)} = \frac{\mu_i}{\lambda_i + \mu_i}. \quad (2)$$

The mean time to failure and mean time to repair are:

$$\text{MTTF}_i = \frac{1}{\lambda_i}, \quad \text{MTTR}_i = \frac{1}{\mu_i}. \quad (3)$$

4.2 Time-Dependent Availability via Matrix Exponential

Starting from state 0 (working) at time $t = 0$, the probability that component i is working at time t is:

$$A_i(t) = [e^{\mathbf{Q}_i t}]_{00} = \frac{\mu_i}{\lambda_i + \mu_i} + \frac{\lambda_i}{\lambda_i + \mu_i} e^{-(\lambda_i + \mu_i)t}. \quad (4)$$

As $t \rightarrow \infty$, $A_i(t) \rightarrow A_i$, consistent with (2). The matrix exponential is computed numerically via `scipy.linalg.expm`.

Illustrative values for Component 1 ($\lambda = 0.002$, $\mu = 0.10$):

Table 2: Time-dependent availability $A_1(t)$ for Component 1.

t (hours)	$A_1(t)$
10	0.9875
100	0.9804
500	0.9804
1000	0.9804

The system reaches its steady state rapidly — within roughly 100 hours — after which $A_1(t)$ remains constant at the steady-state value 0.9804. This fast convergence is characteristic of high-repair-rate systems.

4.3 System Topology Reliability

4.3.1 Series Configuration

In a series system, all n components must work simultaneously. Under the assumption of independence:

$$R_{\text{series}} = \prod_{i=1}^n A_i = 0.630614. \quad (5)$$

This is the most pessimistic configuration. With 15 components each at $\sim 97\%$ availability, even a small per-component failure probability compounds to a 37% system unavailability.

4.3.2 Parallel Configuration

In a fully parallel system, at least one component must work:

$$R_{\text{parallel}} = 1 - \prod_{i=1}^n (1 - A_i) \approx 1.000000. \quad (6)$$

The parallel system achieves near-perfect reliability, as the probability that all 15 components fail simultaneously is vanishingly small.

4.3.3 k -out-of- n Configuration

The k -out-of- $n : G$ model requires at least k of n components to work. Assuming homogeneous availability $p = \bar{A} = \frac{1}{n} \sum_i A_i$:

$$R_{k/n} = \sum_{j=k}^n \binom{n}{j} p^j (1-p)^{n-j}. \quad (7)$$

Table 3: System reliability under different topologies.

Configuration	Reliability
Series (all 15 must work)	0.630614
Parallel (at least 1 works)	1.000000
8-out-of-15	1.000000
10-out-of-15	0.999997
12-out-of-15	0.999641

The high system reliability under k -out-of- n for $k \leq 10$ reflects the strong individual component availabilities. The series system is the binding constraint in fault-intolerant designs.

5 Monte Carlo Simulation

5.1 Simulation Design

For each of $M = 100,000$ independent trials, we simulate all 15 components over a mission horizon $T = 1,000$ hours. Each component i is simulated using the competing-exponentials method:

1. Initialize component in the working state ($s = 0$) at $t = 0$.
2. While $t < T$:
 - If $s = 0$: draw sojourn time $\tau \sim \text{Exp}(\lambda_i)$. Advance $t += \tau$. If $t < T$, transition to $s = 1$ and increment failure counter.
 - If $s = 1$: draw repair time $\rho \sim \text{Exp}(\mu_i)$. Advance $t += \rho$. If $t < T$, transition to $s = 0$.
3. Record working time W_i and failure count N_i .

The per-trial availability estimate for component i is $\hat{A}_i = W_i/T$. System-level availability is computed under the 8-out-of-15 rule: a trial is a “system success” if $\sum_i \mathbf{1}[\hat{A}_i > 0.5] \geq 8$.

5.2 Estimated Metrics

Table 4: Monte Carlo estimates ($M = 100,000$, $T = 1000$ hr).

Metric	MC Estimate	Analytical
Mean component availability	0.9702	0.9699
System availability (8/15)	1.0000	1.0000
Estimated system MTTF	25.32 hr	—
Avg. failures per component	2.63	$\lambda_i T$

5.3 Validation

The MC mean component availability (0.9702) matches the analytical value (0.9699) with absolute error < 0.001 , validating the simulation implementation. The system MTTF of 25.32 hours reflects the expected time between system-level degradation events (at least 8 components failing simultaneously), consistent with the high per-component repair rates.

By the strong law of large numbers, as $M \rightarrow \infty$:

$$\frac{1}{M} \sum_{m=1}^M \hat{A}^{(m)} \xrightarrow{\text{a.s.}} \mathbb{E}[\hat{A}] = \bar{A}.$$

The central limit theorem gives the approximate 95% confidence interval for the mean availability estimate:

$$\bar{A} \pm 1.96 \cdot \frac{s}{\sqrt{M}},$$

where s is the sample standard deviation across trials.

6 Sensitivity Analysis

6.1 Methodology

To quantify the marginal impact of each component's transition rates on system reliability, we compute *analytical sensitivity indices* via finite differencing of the closed-form reliability expression. For a perturbation $\delta = 10\%$:

$$S_{\lambda_i} = \frac{R_{k/n}(\lambda_i(1 + \delta)) - R_{k/n}(\lambda_i)}{\delta \cdot \lambda_i}, \quad (8)$$

$$S_{\mu_i} = \frac{R_{k/n}(\mu_i(1 + \delta)) - R_{k/n}(\mu_i)}{\delta \cdot \mu_i}. \quad (9)$$

Using closed-form availability (2) and reliability (7) avoids Monte Carlo noise entirely,

yielding exact sensitivity indices.

6.2 Results and Interpretation

All sensitivity indices are of order 10^{-6} , which is a genuine mathematical result rather than a numerical artifact. This arises because the 8-out-of-15 system reliability is essentially 1.0 at baseline: the function $R_{8/15}(\cdot)$ is extremely flat in the neighborhood of the current operating point, and perturbations to any single component have negligible marginal effect on the already near-perfect system reliability.

This finding has important practical implications:

- **System robustness:** The 8-out-of-15 architecture is inherently robust to individual component degradation. No single component is a bottleneck.
- **Sensitivity shifts under stricter requirements:** If the threshold were raised to $k = 13$ or 14 , individual components would become critical and sensitivities would be substantially larger.
- **Series system contrast:** The series topology ($R = 0.631$) sits on a steep portion of the reliability curve, where sensitivities are much larger. Components 6 and 13 ($\lambda = 0.005$, the highest failure rates) would be identified as the primary risk drivers.

7 Importance Sampling for Rare-Event Estimation

7.1 Motivation

Standard Monte Carlo is inefficient for estimating probabilities $p = P(A)$ when $p \ll 1$. To achieve a relative error of ε , one requires approximately $O(1/(\varepsilon^2 p))$ samples. For $p \sim 10^{-18}$, this is computationally infeasible.

7.2 Importance Sampling Estimator

Let f denote the nominal density (failure rates λ_i) and g the importance (biased) density (failure rates $\lambda_i^{\text{IS}} = \beta \lambda_i$ with inflation factor $\beta = 5$). The IS estimator of $p = \mathbb{E}_f[\mathbf{1}_A(X)]$ is:

$$\hat{p}_{\text{IS}} = \frac{1}{M} \sum_{m=1}^M \mathbf{1}_A(X^{(m)}) w(X^{(m)}), \quad w(X) = \frac{f(X)}{g(X)}, \quad (10)$$

where $X^{(m)} \sim g$ and $w(X)$ is the likelihood ratio weight.

7.3 Likelihood Ratio for Poisson Failure Counts

Under the Poisson approximation (valid since $\lambda_i T \leq 5$ for all components), the number of failures $N_i \sim \text{Pois}(\lambda_i T)$ under f and $N_i \sim \text{Pois}(\lambda_i^{\text{IS}} T)$ under g . The log-likelihood ratio is:

$$\log w = \sum_{i=1}^n \left[(\lambda_i^{\text{IS}} - \lambda_i) T + N_i \log \left(\frac{\lambda_i T}{\lambda_i^{\text{IS}} T} \right) \right] = \sum_{i=1}^n \left[(\beta - 1) \lambda_i T \cdot \frac{\beta - 1}{\beta} - N_i \log \beta \right]. \quad (11)$$

7.4 Results

We estimate $p = P(\geq 5 \text{ components fail } \geq 1 \text{ time in } 1000 \text{ hr})$ using $M = 50,000$ IS trials:

Table 5: Importance sampling results for rare-event estimation.

Quantity	Value
IS estimate $\hat{p}$	4.25×10^{-18}
Standard error	1.03×10^{-35}
95% CI	$(4.25 \times 10^{-18}, 4.25 \times 10^{-18})$
Inflation factor β	5
Number of IS samples	50,000

The extremely small probability ($\sim 10^{-18}$) reflects the high individual component reliabilities: even inflating failure rates by $5\times$, simultaneous failure of 5 or more components within 1000 hours is astronomically rare.

8 Redundancy Optimization

8.1 Problem Formulation

We seek to allocate redundant components across the 15 subsystems to maximize series-of-parallel-subsystems reliability subject to a total cost budget B :

$$\begin{aligned} & \max_{r_1, \dots, r_n \in \mathbb{Z}_{\geq 0}} \prod_{i=1}^n [1 - (1 - A_i)^{1+r_i}] \\ & \text{subject to } \sum_{i=1}^n c_i r_i \leq B, \\ & 0 \leq r_i \leq r_{\max}, \quad i = 1, \dots, n, \end{aligned} \quad (12)$$

where r_i is the number of additional redundant copies of component i , c_i is the unit cost, $B = \$5,000$, and $r_{\max} = 3$.

8.2 Greedy Marginal-Gain Algorithm

Problem (12) is a nonlinear integer program. We solve it with a greedy algorithm that, at each step, selects the component yielding the highest reliability gain per dollar:

$$i^* = \arg \max_{i: r_i < r_{\max}, c_i \leq B_{\text{rem}}} \frac{\Delta R_i(r_i)}{c_i}, \quad (13)$$

where $\Delta R_i(r_i) = [1 - (1 - A_i)^{r_i+2}] - [1 - (1 - A_i)^{r_i+1}]$ is the marginal reliability gain from the next redundant copy of component i . The greedy approach is optimal for submodular functions and provides a $(1 - 1/e) \approx 63\%$ approximation guarantee in general.

8.3 Component Costs and Optimal Allocation

Table 6: Component costs and optimal redundancy allocation.

Component	Unit cost (\$)	Added units	Cost allocated (\$)	Subsystem reliability
1	500	0	—	0.99962
2	800	1	800	0.99869
3	300	1	300	0.99993
4	1000	0	—	0.99824
5	600	0	—	0.99966
6	900	1	900	0.99956
7	700	0	—	0.99912
8	500	0	—	0.99952
9	400	0	—	0.99994
10	800	1	800	0.99773
11	600	1	600	0.99991
12	500	0	—	0.99997
13	900	1	900	0.99956
14	350	0	—	0.99993
15	700	1	700	0.99899
Total	—	7 units	\$5,000	—

8.4 Results

Table 7: Redundancy optimization outcome.

Metric	Value
Baseline series reliability	0.630614
Optimized series reliability	0.834211
Absolute improvement	+0.203597
Relative improvement	+32.3%
Budget utilized	\$5,000 / \$5,000

The optimizer allocates redundancy to components 2, 3, 6, 10, 11, 13, and 15 — precisely those with the most unfavorable failure-rate-to-cost ratios. The 20.4 percentage-point improvement in absolute reliability (from 0.631 to 0.834) represents a significant reduction in system unavailability of 55%.

9 Applications of CTMC Reliability Analysis

The methodology developed in this project has broad applicability across engineering, science, and finance. Below we detail the most significant application domains.

9.1 Nuclear Power and Safety-Critical Systems

Nuclear power plants employ redundant safety systems (emergency cooling, control rod actuation, containment) designed to meet extremely stringent reliability requirements (typically $R > 1 - 10^{-6}$ per demand). CTMCs model the transition between normal operation, degraded states, and shutdown [5]. Importance sampling is indispensable for estimating core damage frequencies on the order of 10^{-5} – 10^{-7} per reactor-year, which are entirely inaccessible to standard simulation.

9.2 Telecommunications and Network Infrastructure

Internet backbone routers, 5G base stations, and submarine cable systems operate as k -out-of- n configurations. Service providers use CTMC-based availability models to meet Service Level Agreements (SLAs) guaranteeing 99.999% (“five nines”) uptime. Redundancy optimization directly informs infrastructure investment decisions worth billions of dollars annually.

9.3 Aerospace and Aviation

Aircraft hydraulic, electrical, and flight control systems must demonstrate reliability in excess of $1 - 10^{-9}$ per flight hour. The FAA and EASA require probabilistic safety as-

assessments (PSAs) based on fault trees and Markov models. Monte Carlo simulation with variance reduction is used to verify that catastrophic failure modes meet the extremely low probability thresholds mandated by certification standards.

9.4 Power Grids and Smart Energy Systems

Electric utility companies use CTMC reliability models to plan maintenance schedules, quantify loss-of-load probability (LOLP), and evaluate the benefit of adding generation or transmission capacity. The North American Electric Reliability Corporation (NERC) mandates probabilistic reliability assessments using exactly the class of models developed here.

9.5 Financial Risk and Operational Resilience

In quantitative finance, *credit migration models* (e.g., Moodys/S&P rating transitions) are CTMCs where states correspond to credit ratings (AAA, AA, ..., D). Portfolio-level default probabilities and CVA (Credit Valuation Adjustment) are estimated via Monte Carlo simulation on these chains [6]. Importance sampling is used to estimate tail probabilities (e.g., probability of a 10-firm simultaneous default) for stress testing under Basel III/IV requirements.

9.6 Healthcare and Medical Device Reliability

Implantable medical devices (pacemakers, insulin pumps, deep brain stimulators) must demonstrate extraordinary reliability over 5–10 year deployment horizons. CTMC models capture battery depletion, firmware states, and component wear. Redundancy optimization informs component selection in FDA submissions.

9.7 Software and Cloud Computing Systems

Microservice architectures, container orchestration platforms (Kubernetes), and cloud storage systems (e.g., Amazon S3's 11-nines durability guarantee) rely on k -out-of- n replication across availability zones. CTMC-based models estimate the probability of data loss under correlated failure events, informing replication strategies and SLA design.

9.8 Manufacturing and Supply Chain

Production lines model machine states (running, broken down, under maintenance) as CTMCs. Overall Equipment Effectiveness (OEE) is directly related to steady-state availability A_i . Redundancy allocation (spare parts stocking, backup machines) under budget constraints is solved using exactly the formulation in Section ??.

10 Discussion

10.1 Key Findings

1. **High baseline reliability:** With individual component availabilities ranging from 93.3% to 99.2%, the 8-out-of-15 system achieves effectively 100% availability under all analytical and simulation-based metrics. This confirms that fault-tolerant majority-voting architectures are extremely robust for this parameter regime.
2. **Series topology is the risk driver:** The series configuration yields $R = 0.631$, meaning over one-third of operating time involves at least one component failure. In fault-intolerant designs, components 6 and 13 (highest λ , lowest availability at 93.3%) would be priority targets for reliability improvement.
3. **Sensitivity flatness validates architecture:** The uniformly near-zero sensitivity indices confirm that the 8-out-of-15 system is operating in a regime where no individual component is a systemic risk factor. This is architecturally desirable and provides confidence that the design is robust to component-level uncertainty.
4. **Redundancy provides substantial gains:** The greedy optimizer achieves a 32% relative improvement in series reliability by allocating all \$5,000 to the seven highest-value components. The full budget utilization confirms the optimizer is not leaving value on the table.
5. **Rapid convergence to steady state:** Time-dependent availability reaches its steady-state value within approximately 100 hours for all components, implying that transient startup effects are negligible for mission horizons of 1,000 hours.

10.2 Limitations

- The model assumes exponentially distributed sojourn times (memoryless property). Real components often exhibit wear-out failure modes better described by Weibull or log-normal distributions.
- Component failures are treated as independent. Correlated failures due to shared environments (common-cause failures) are not modeled but are important in practice.
- The k -out-of- n model assumes a homogeneous threshold. Weighted voting or hierarchical topologies would require a more complex system reliability model.
- The Poisson approximation used in importance sampling is valid only when $\lambda_i T$ is not too large. For high failure-rate or long-mission scenarios, exact sojourn simulation should be used.

10.3 Future Directions

- Extend to non-exponential sojourn distributions using phase-type approximations or semi-Markov processes.
- Incorporate common-cause failure groups into the CTMC model.
- Apply multi-level Monte Carlo (MLMC) for variance reduction with provable convergence rates.
- Solve the integer redundancy optimization exactly using branch-and-bound or dynamic programming.
- Calibrate λ_i and μ_i from real failure data using Bayesian hierarchical models (natural extension of existing Bayesian capstone work).

11 Conclusion

This project has developed and implemented a comprehensive stochastic reliability analysis framework for a 15-component networked system. Using Continuous-Time Markov Chains as the theoretical foundation, we derived exact closed-form results for component and system-level reliability metrics, validated these against large-scale Monte Carlo simulation with 10^5 trials, and demonstrated that simulation and theory agree to within 0.03%.

The 8-out-of-15 fault-tolerant architecture was shown to be highly robust, achieving near-perfect system availability and exhibiting negligible sensitivity to individual component parameter variations. The series topology, by contrast, was identified as the critical vulnerability, with reliability of only 63%.

Importance sampling successfully estimated rare-event probabilities on the order of 10^{-18} — quantities utterly inaccessible to standard Monte Carlo — while the greedy redundancy optimizer achieved a 32% relative improvement in system reliability within a \$5,000 budget by intelligently targeting the highest-value components.

The methodology presented here is directly applicable across nuclear engineering, telecommunications, aerospace, power systems, finance, healthcare, and manufacturing — any domain where quantifying and improving the reliability of complex interconnected systems is of practical importance. The clean separation between analytical and simulation-based approaches, and the use of variance reduction techniques for rare-event estimation, make this framework a strong foundation for both research and engineering practice.

Acknowledgments

The author acknowledges the Department of Mathematics and Statistics at Georgia State University for providing the computational and academic resources that supported this work. Computational experiments were performed using Python 3.11 with NumPy 1.26, SciPy 1.12, and pandas 2.2.

References

References

- [1] Barlow, R. E., & Proschan, F. (1965). *Mathematical Theory of Reliability*. John Wiley & Sons, New York.
- [2] Norris, J. R. (1997). *Markov Chains*. Cambridge University Press, Cambridge.
- [3] Glynn, P. W., & Iglehart, D. L. (1989). Importance sampling for stochastic simulations. *Management Science*, 35(11), 1367–1392.
- [4] Kuo, W., & Wan, R. (2003). Recent advances in optimal reliability allocation. *IEEE Transactions on Systems, Man, and Cybernetics*, 37(2), 143–156.
- [5] Bedford, T., & Cooke, R. (2001). *Probabilistic Risk Analysis: Foundations and Methods*. Cambridge University Press, Cambridge.
- [6] Lando, D. (2004). *Credit Risk Modeling: Theory and Applications*. Princeton University Press, Princeton.
- [7] Department of Defense (1991). *MIL-HDBK-217F: Reliability Prediction of Electronic Equipment*. US Department of Defense, Washington, D.C.
- [8] Trivedi, K. S. (2002). *Probability and Statistics with Reliability, Queuing, and Computer Science Applications* (2nd ed.). John Wiley & Sons, New York.
- [9] Rubinstein, R. Y., & Kroese, D. P. (2016). *Simulation and the Monte Carlo Method* (3rd ed.). John Wiley & Sons, Hoboken, NJ.
- [10] Lisnianski, A., & Levitin, G. (2003). *Multi-State System Reliability: Assessment, Optimization and Applications*. World Scientific, Singapore.

A Python Implementation Notes

A.1 Key Library Functions

- **Matrix exponential:** `scipy.linalg.expm(Q * t)` computes e^{Q^t} using Padé approximation with scaling and squaring.
- **Random variate generation:** `numpy.random.default_rng(seed)` initializes a PCG64 generator; `rng.exponential(scale)` draws from $\text{Exp}(1/\text{scale})$.
- **Optimization:** The greedy redundancy allocator runs in $O(n \cdot B/c_{\min})$ time, which is computationally feasible for all realistic parameter values.

A.2 Reproducibility

All results are fully reproducible with `RANDOM_SEED = 42`. The complete codebase is self-contained in a single Python file (`ctmc_reliability.py`) with no external data dependencies.

A.3 Computational Cost

Table 8: Approximate runtime for each analysis component.

Component	Complexity	Runtime
Analytical metrics	$O(n)$	< 1 ms
CTMC matrix exponential	$O(\mathcal{S} ^3)$	< 1 ms
Monte Carlo (10^5)	$O(M \cdot n \cdot T\bar{\lambda})$	~30 s
Sensitivity analysis	$O(n)$	< 1 ms
Importance sampling	$O(M_{\text{IS}} \cdot n)$	~60 s
Redundancy optimization	$O(n \cdot B/c_{\min})$	< 1 ms